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22 August 1986
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MILITARY SPECIFICATION

GRINDING MACHINES, SURFACE, ROTARY, VERTICAL SPINDLE, SINGLE TABLE, MANUAL AND AUTOMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, vertical spindle, single rotary table, manual or power operated surface grinding machines with equipment and accessories specified herein.

1.2 Classification. The grinding machines shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Manually operated.
Type II - Automatic cycle.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Size	Grinding wheel diameter (inches)	Chuck or worktable diameter (inches)
812 +	8	12/14*
1116	11	16/20*
2036	20	36/42*
2242	22	42
2648	26/28*	48/54*
3260	32	60
3666	36	60/66/72*
4272 +	42/44*	72/84*
4896	48	96
54100 +	54	100/108*
60120	60	120
72144 +	72	120/144*
80160	80	160

+ Sizes 812, 4284, 54100, and 72120 available in type I only; Sizes 4272, 4484, 54108, and 72144 available in type I and type II.

* Optional sizes for both grinding wheel diameter and chuck or worktable diameter available as indicated.

2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

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MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NEMA ICS 6 - Enclosures for Industrial Controls and Systems.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B7.1 - Safety Requirements for the Use, Care and Protection of Abrasive Wheels.

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines.

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels and Shapes, Sizes, and Identification of Mounted Wheels.

ANSI B74.13 - Specifications for Marking for Identifying Grinding Wheels and Shapes, Sizes, and Identification of Mounted Wheels.

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(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models. Type I machines shall be manually operated requiring the

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operator to perform all machine functions. Type II machines shall be automatic, performing all machine functions from positioning the chuck and starting up the grinding wheel to stopping the wheel and demagnetizing the chuck, after the operator magnetizes the chuck and performs set-up and machine start-up duties. Each machine shall be capable of heavy stock removal and precision finishing of glass, quartz, silicon, ferrous, and nonferrous workpieces in accordance with 3.6. Each machine shall be designed for cylinder or segmental wheel grinding using the wheel face to contact workpieces. Machine design shall include all components, parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2.1), a digital readout system shall be provided to reference the wheelhead position (Z and X axes). The digital readout system shall consist of an electronic display for each axis and all related components, mounting hardware, and electrical equipment necessary for complete operation of the system. The system shall be capable of providing a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.00004 inch. The system accuracy shall be 0.0002 inch per foot and the repeatability shall be 0.00010 inch. Special requirements for digital readout system components shall be as specified (see 6.2.1). The readout system shall be capable of indicating axis location in both the US and SI systems of measurement.

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3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Operator's controls. Type I and type II machines shall be provided with an oil-tight overhead pendent or push button station located convenient to the normal operator position and, unless otherwise specified (see 6.2.1), shall have at least the following features:

Grinding wheel start/stop (rotation)
Table or carriage in/out/jog
Chuck start/stop (rotation)
Grinding head up/down/jog
Coolant start/stop
Emergency stop (entire machine)
Chuck on/off

Type II machines shall also have a "Cycle start/stop" feature (see 3.4.7).

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the grinding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of ANSI B11.9 and OSHA 29 CFR that are applicable to the machine itself. In the event of a conflict between the requirements of ANSI B11.9 and OSHA 29 CFR, the requirements of OSHA 29 CFR shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous

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to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent loosing. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.3.8 Machine ways. Machine guideways shall be adequately protected regardless of machine component position to exclude chips, dust, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members.

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3.3.9 Slides. All sliding members shall be provided with gibs, guides, or other means to maintain the alignment and accuracy tolerances required herein for the machine type specified.

3.3.10 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.11 Electromagnetic interference control. When specified (see 4.7.4 and 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Each machine shall consist of not less than a base, column, table, magnetic chuck, grinding wheel and grinding wheelhead assembly, wheel dresser, drive mechanisms, and controls.

3.4.1 Base The base shall be a casting of iron or iron alloy or a fabricated steel weldment, ribbed and braced to minimize distortion. Longitudinal guideways shall be provided to accept a traversing table or carriage. Guideways shall support the loaded table or carriage and magnetic chuck from the loading position to the grinding position. Means shall be provided to accept the column and its related components. Means shall be provided to level and anchor the machine to the floor. Hydraulic and coolant reservoirs, when provided, shall be separate external units. When applicable, an access door(s) or removable plate(s) shall be provided for servicing components housed within the base. Means shall be provided to collect and discharge coolant and grinding swarf from the base. When specified (see 6.2.1), an automatic chip collection system with provisions for chip and sludge removal shall be provided separate from the base.

3.4.2 Column. The column shall be a one-piece casting, ribbed and braced to minimize distortion and deflection. The surfaces of the column shall accurately fit mating surfaces of the base and shall be bolted to the base forming single unit construction of the head, column and base. The column shall accommodate a vertically traveling grinding wheelhead assembly, speed and feed components, driving mechanisms, wheel dresser and controls. Means shall be provided to align the column to the work holding surface. An access door(s) or removable plate(s) shall be provided to service housed components, as applicable.

3.4.3 Worktable. The worktable shall be composed of a circular table mounted directly on the base slides or a circular table mounted on a carriage which shall mate with the base slides. The table or carriage shall

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traverse longitudinally along the base slides. Unless otherwise specified (see 6.2.1), type I and type II machines shall have power table traverse. Means shall be provided to prevent movement of the table or carriage after positioning. An independent drive unit and controls shall be provided on the table for table traverse. When specified (see 6.2.1), an extra table the same diameter as the work table shall be provided to mount on the work table to be used for grinding with fixtures in place of a magnetic chuck. The extra worktable shall be drilled or T-slotted, as specified (see 6.2.1). The length of table or carriage travel and the drive mechanism requirements shall be as specified in table I.

3.4.4 Magnetic chuck. The chuck shall be a rotating magnetic chuck securely mounted on the worktable. The chuck shall be properly aligned with the column and wheelhead to assure compliance with the flatness grinding requirements specified in 3.6. The chuck shall be mounted on antifriction bearings or machined, ground and scraped circular cast iron flat ways and provided with proper lubrication. An independent drive unit shall be furnished on the chuck to provide rotary motion of the workpiece. The chuck body shall be steel, machined, and fitted with coils and strips adequately insulated and sealed in the face of the chuck to assure no fluid leakage into the coils or accumulation of moisture therein. The chuck shall securely hold the workpiece on the chuck face under maximum metal removal conditions in accordance with the manufacturer's recommended loading and blocking procedures. The poles of the magnetic chuck shall be not more than 1-1/16 inches apart over the entire face of the chuck so small parts as well as large parts can be held securely during grinding. On all machines larger than the size 812, means shall be provided to demagnetize the chuck to allow for workpiece loading and unloading. Chuck size, motor, and rotation requirements shall be as specified in table I for the machine type and size specified.

3.4.5 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a wheelhead housing, spindle, spindle drive unit, and a drive unit(s) for wheelhead assembly vertical positioning and grinding wheel downfeed. The spindle drive unit shall operate at a speed necessary to provide a single grinding wheel peripheral speed within the range of 3400 to 5000 surface feet per minute (SFPM). The grinding spindle shall be mounted in antifriction bearings designed to minimize axial and radial play and shall be dynamically balanced. The spindle shall accept a driving mechanism at one end and a grinding wheel and holder at the other end. Continuous or permanent lubrication to the spindle shall be provided. The assembly shall be mounted on the guideways provided on the column and shall traverse vertically. Manual and power rapid raising and lowering of the wheelhead assembly for setup shall be provided. Downfeed of the wheelhead assembly shall be controlled by the operator on type I machines and controlled by the automatic cycle (see 3.4.7) on type II machines. A graduted indicator or digital readout shall be provided to monitor grinding wheelhead assembly position. When specified (see 6.2.1), the machine shall be provided with a power-actuated tilting column or wheelhead assembly. When the machine is equipped with a tilting column or wheelhead assembly, a single control shall be provided to actuate the tilting mechanism and return the mechanism to a

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zero reference point. When specified (see 6.2.1), the grinding wheel shall be provided with dynamic braking to assure rapid and smooth stopping of the wheel.

3.4.5.1 Manual grinding wheel downfeed. Manual downfeed of the grinding wheel into the work shall be accomplished utilizing a handwheel or push button. Both coarse and fine feed shall be available with a graduated indicator or digital readout provided to monitor wheel position.

3.4.5.2 Automatic grinding wheel downfeed. Each type II machine shall be equipped with automatic downfeed of the grinding wheel. Means shall be provided for controlling feed and limiting the total downfeed for the overall grinding cycle. Downfeed rates and ranges shall be as specified in table I.

3.4.6 Automatic sizing. When specified (see 6.2.1), type I machines shall be provided with an automatic sizing system. The system shall give continuous measurement of the work during grinding and visual indication the work is approaching size. When the work reaches a pre-set dimension, feed shall stop, the machine shall spark out, and the wheel shall retract.

3.4.7 Automatic cycle. Type II machines shall be provided with automatic cycle. After manually inputting predetermined machine and workpiece parameters, loading the workpiece, and activating the magnetic chuck, the operator shall complete a grind cycle by starting the automatic cycle from a single control. The automatic cycle shall include, but not be limited to, the following functions:

a. Moving the chuck to the grinding position.
b. Starting chuck rotation.
c. Starting grinding wheel rotation.
d. Engaging coolant pump.
e. Engaging power downfeed at pre-set rate until stopped by automatic gauging system, including control of the head tilt feature, when applicable.
f. Accomplishing pre-set timed sparkout.
g. Retracting wheelhead to pre-set location.
h. Stopping the grinding wheel rotation and coolant pump.
i. Retracting the chuck to the load position, stopping chuck rotation, and demagnetizing the chuck.

If the optional fixture table is used (see 3.4.3), the automatic cycle shall be the same without chuck actuation and interlock functions. The automatic cycle shall be capable of being interrupted and re-entered during running.

3.4.8 Drive mechanisms. Unless otherwise specified (see 6.2.1), the following drive mechanisms shall position each moving member in all axes of motion. Manual positioning shall be accomplished utilizing handwheels or pushbuttons and precision leadscrews or ballscrews; movement shall be initiated and controlled by the operator. Power movement shall be accomplished

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by solid-state servo drives, electric motors, and precision leadscrews or precision, preloaded, antifriction recirculating ballscrew and nut assemblies. Leadscrews or ballscrews shall be hardened and precision ground to provide alignment and positioning accuracy, and be of a size, class, and precision suitable for the application. Leadscrews or ball screws and nuts shall not exert excessive loads or cause premature deterioration of the feed drive motors or components.

3.4.9 Wheel dresser. A permanently mounted wheel dresser capable of dressing the entire working face of both solid cylindrical and segmented grinding wheels shall be provided on each machine. The dresser shall be manually operated.

3.4.10 Grinding wheels. Unless otherwise specified (see 6.2.1), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All wheels shall be properly guarded.

3.4.11 Lubrication system. A lubrication system shall be provided to ensure adequate lubrication for all moving parts. When specified (see 6.2.1), the lubrication system shall be completely automatic. The system shall deliver lubricant to all mating gears, sliding parts and bearings, except the sealed-for-life type. Each lubricant reservoir shall have a device for indicating and maintaining fluid level. The lubricant reservoir shall operate for a minimum of 72 hours of feed time without being serviced. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Interlocks shall be provided on the wheelhead and the table or carriage to prevent operation in case of pressure loss.

3.4.12 Coolant system. A coolant system shall be provided on each machine. The coolant system shall be either mist or flood type, as specified (see 6.2.1). A coolant delivery nozzle(s) and valve(s) shall permit the operator to direct and control the amount of coolant flow to the work and flush the base. When specified (see 6.2.1), all coolant piping and valves required for directing coolant to the inside of the grinding wheel shall be provided. The flood type system shall include a sump or reservoir, a power driven pump, and piping. The mist type system shall include a reservoir, filter, means for pressurizing the coolant, and means for directing and controlling the coolant. The coolant reservoir shall be of sufficient capacity to permit full-flow coolant. The coolant system shall have means for draining and shall include a baffle or strainer, easily removable for cleaning. The reservoir shall have means for emptying and cleaning. Type I machine coolant systems shall be manually controlled and type II machine coolant systems shall be controlled by the automatic cycle (see 3.4.7).

3.4.13 Hydraulic system. If a hydraulic system is provided, it shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system

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shall ensure delivery of clean fluid to the hydraulic system. The hydraulic reservoir shall have a sight gauge to indicate the fluid level. The fluid temperature shall be maintained within the proper operating range during all normal operations of the machine. Means shall be provided to drain and clean out the hydraulic reservoir.

3.4.14 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt (with an allowable line fluctuation of ±10 percent), 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. Each machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating. Control enclosures shall meet the requirements of ANSI/NEMA ICS 6.

3.4.14.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG 1.

3.4.14.2 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary nominal voltage of 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine type and size specified.

3.6 Performance. Each machine shall be capable of removing 1/4 cubic inch of cast iron per spindle motor rated horsepower per minute for roughing operations. Test surfaces shall be finish-ground to not rougher than 40 microinch. For machine sizes up to 84 inches in diameter, finished test blocks shall be parallel, top and bottom, and of the same thickness within 0.0002 inch per foot of their separation on the chuck. For machine sizes 96 inches in diameter and larger, finished test blocks shall be parallel, top and bottom, and of the same thickness within 0.0004 inch per foot of their separation on the chuck. Machine performance shall be tested in accordance with the test procedures given in 4.7.1 and 4.7.2.

3.6.1 Additional performance requirements. Performance requirements in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances of 4.7.3.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of special wrenches normally furnished by the manufacturer for the machine.

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b. Grinding wheels: Number of each size, type, dimensional characteristics and composition shall be as specified (see 6.2.1), with required number of wheel holders, as applicable.

c. Table splash guards.

d. One chuck ring, outside, to prevent small work (60 inches in diameter or less) from sliding.

3.9 Optional equipment. Optional equipment and accessories shall be furnished as specified and described (see 6.2.1).

3.10 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface letters on a contrasting background.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a corrosion-resistant lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.12 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments

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of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.13 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. Each item must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated test area under normal environmental conditions.

4.4 First article inspection. When a first article inspection is required it shall be applied to the first article submitted in accordance

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with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6, all tests in 4.7, and the inspection in 4.8. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1 through 4.7.3, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The machine and equipment shall be examined for compliance with the requirements of this specification in 3.2 through 3.5 and 3.8 through 3.14, including compliance with all referenced standards.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 6 hours. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be verified during the trial period.

4.7.2 Performance tests. The machine shall be tested to determine its ability to rough and finish grind both top and bottom surfaces of eight test blocks, four placed approximately 6 inches in from the outer periphery of the magnetic chuck, spaced equally apart, and four placed approximately 6 inches from the inner face of the magnetic chuck, spaced equally apart. The test blocks shall be of cast iron, size to be determined by the supplier dependent upon the size of machine purchased. The machine shall meet the performance requirements given in 3.6. The methods used to obtain the above designated requirements shall be at the supplier's option. At the option of the procuring activity, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the described test (see 6.2.1).

4.7.3 Alignment tests. Each machine and its components shall be tested for alignment. Unless otherwise specified (see 6.2.1), the alignment requirements shall be in accordance with table II. The instruments and methods used in performing the tests shall be at the option of the supplier, provided they accurately determine the tolerances of the machine.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.11.

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4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in shops for rough and finish grinding of workpieces with rapid metal removal rates to produce precision flat finishes on a wide variety of materials.

a. Title, number and date of this specification.

b. Type and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Digital readout system and any special requirements, if required (see 3.2.1.2).

g. Operator's controls, if different (see 3.2.4).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.11).

k. Automatic chip removal system, if required (see 3.4.1).

l. Traverse on table or carriage, if different (see 3.4.3).

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m. Extra work table, if required; specify drilled or T-slotted table, as required (see 3.4.3).

n. Power-actuated tilting column or wheelhead assembly, if required (see 3.4.5).

o. Dynamic grinding wheel braking, if required (see 3.4.5).

p. Automatic sizing system on type I machine, if required (see 3.4.6).

q. Drive mechanisms, if different (see 3.4.8).

r. Grinding wheels, if different (see 3.4.10).

s. Automatic lubrication system, if required (see 3.4.11).

t. Specify type of coolant system required; coolant inside grinding wheel, if required (see 3.4.12).

u. Electrical system and input power, if different (see 3.4.14).

v. Motor enclosure, if different (see 3.4.14.1).

w. Size and capacity, if different (see 3.5).

x. Additional performance requirements, if required (see 3.6.1).

y. Standard equipment, if different (see 3.8).

z. Grinding wheel number, type, dimensional characteristics, composition, and required number of wheel holders (see 3.8.b).

aa. Specify and fully describe optional equipment, if required (see 3.9).

bb. Specify applicable repair parts, if required (see 3.10).

cc. Lubrication plate, if different (see 3.11.1).

dd. Nameplate, if different (see 3.11.2).

ee. Hour meter, if required (see 3.12).

ff. Inspection conditions, if different (see 4.3).

gg. First article inspection, if different (see 4.4).

hh. Quality conformance inspection, if different (see 4.5).

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ii. Additional performance test(s) as fully described, if required (see 4.7.2).

jj. Alignment requirements, if different (see 4.7.3).

kk. If preservation, packing and marking in accordance with MIL-M-18058 are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern the "machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional requirements that will integrate the new machine into its future operating environment.

6.5 Cross reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-G-80125A	MIL-G-80125B	
----	Types I and II, Size 812	New
Size 1116	Size 1116	
Size 2036	Size 2036	
Size 2242	Size 2242	
Size 2648	Size 2648	
Size 3260	Size 3260	
Size 3660	Size 3666	
Size 4272	Size 4272	
Size 4896	Size 4896	
Size 54100	Size 54100	
Size 60120	Size 80160	
----	Size 72144	New
----	Size 80160	New

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6.6 Subject term (key word) listing:

Abrasive wheel, grinding
Automatic cycle
Grinder, surface
Grinding machine, surface
Machine tool, grinding
Magnetic chuck
Rotary table, traveling, single
Surface grinder
Vertical spindle, single
Wheel, abrasive, grinding

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristics	Requirements					
	Type I and type II					
	Size 812 +	Size 1116	Size 2036	Size 2242	Size 2648	Size 3260
Capacities:						
Chuck or table diameter, inches	12/14*	16/20*	36/42*	42	48/54*	60
Swing, inside splash guards, inches	14	20	38	46	52	61
Distance, face of chuck to new wheel, inches	6	7	12	23	23	24
Grinding wheel diameter, inches	8	11	20	22	26/28*	32
Table traverse, inches	8	14	24	27	30	36
Feeds and speeds:						
Wheelhead, power feeds, number (min)	< Infinitely variable within the range >					
Wheelhead feed range, ipm	.001 to .050	.005 to .070	<		.005 to .080	>
Chuck speed, number (min)	variable	variable	variable	variable	< 4	>
Chuck speed, range, rpm	20 to 120	5-1/2 to 20-1/2	8 to 24	8 to 24	8 to 32	7 to 20-1/2
Spindle speed, rpm	1800	1200	720	720	650	460
Motors, HP:						
Spindle	3	15	50	50	75	100
Chuck, rotary	1/2	1	2	3	5	3
Chuck, traverse	manual	1	2	2	3	3
Wheelhead, traverse	1/2	1/2	1	1	2	2
Wheelhead, feed	1/4	1/4	1/4	1/4	1/4	1/2
Coolant	1/6	1/4	1/2	1/2	1/2	3/4

* Indicates optional sizes available.
\+ Sizes 812, 4284, 54100, and 72120 available in type I only.

NOTE Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE I. Sizes and capacities - continued.

Characteristics	Requirements						
	Type I and type II						
	Size 3666	Size 4272+	Size 4896	Size 54100+	Size 60120	Size 72144+	Size 80160
Capacities:							
Chuck or table diameter, inches	60/66/72*	72/84*	96	100/108*	120	120/144*	160
Swing, inside splash guards, inches	84	85	100	125	144	190	190
Distance, face of chuck to new wheel, inches	24	24	30	30	30	48	48
Grinding wheel diameter, inches	36	42/44*	48	70	60	80	80
Table traverse, inches	37	42	54	73	73	78	78
Feeds and speeds:							
Wheelhead, power feeds, number (min)	< ——— Infinitely variable —— within the range —— ——— >						
Wheelhead feed range, ipm	.005 to .080	< ——— .008 to .080 ——— >				< -.005 to .165 - >	
Chuck speed, number (min)	<——— 4 ——— >					< —— variable - >	
Chuck speed, range, rpm	3-1/2 to 22	3 to 11-7/8	3 to 10	2-1/2 to 10	2-1/2 to 10	1 to 10	1 to 10
Spindle speed, rpm	450	337	< ——— 268 ——— >			290	290
Motors, HP:							
Spindle	150	150	150	250	250	250	250
Chuck, rotary	7-1/2	7-1/2	10	10	15	40	40
Chuck, traverse	5	5	5	5	5	7-1/2	7-1/2
Wheelhead, traverse	3	3	3	3	3	3	3
Wheelhead, feed	1/2	1/2	1/2	1/2	1/2	1/2	1/2
Coolant	1	1	3	3	3	5	5

* Indicates optional sizes available.
\+ Sizes 812, 4284, 54100, and 72120 available in type I only.

NOTE Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Alignment test requirements.

Test to be applied	Allowable deviation
Worktable or chuck flat in all directions through center - up to 60" diameter	0.0001/12"
- 72" to 160" diameter	0.0002/12"
Axial runout of worktable or chuck - up to 60" diameter	0.0002" TIR
- 72 to 160" diameter	0.0006" TIR
Surface of worktable or chuck parallel with its transverse movement	0.002" total
Taper of wheel spindle true running	0.0002" TIR
Wheel spindle axial runout - up to 60" diameter	0.0002" TIR
- 72 to 160" diameter	0.0005" TIR
Wheel spindle square with table	0.0015/12"
Vertical adjustment of wheel spindle housing square with table in all planes	0.0015/12"

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Navy - MC, AS, OS

Preparing activity:
DLA - IP

Project (3415-0157)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80125B	Grinding Machines, Surface, Rotary, Vertical Spindle, Single Table, Manual and Automatic

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify): _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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